Registration

Speaker 1

[00:44]
Hello, my name is Yasmin Ferrera, and this presentation is about my MSc Artificial Intelligence project, ScholarPlan.
ScholarPlan is an LLM-powered planning agent designed to support academic research and information gathering. The main idea is that, instead of simply asking an LLM a question and receiving an answer, the system can autonomously plan the research, retrieve academic sources, process the evidence, verify the generated claims, and re-plan when more evidence is needed.
In this presentation, I will explain the motivation behind the project, its architecture and implementation, how the autonomous workflow operates, and the results from testing and evaluation.

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The problem I wanted to address was that academic research involves several different stages. It requires finding relevant sources, managing a large amount of information, turning that evidence into structured findings, and checking whether generated claims are actually supported. If the evidence is insufficient, the research process also needs to be repeated or adapted.
Traditional LLM workflows can produce useful answers, but they don't necessarily provide structured planning, evidence-grounded retrieval, independent verification or adaptive re-planning.
ScholarPlan was designed to address these limitations. It combines LLM-based planning, academic retrieval, claim verification and autonomous re-planning into one bounded research workflow.
So, instead of simply going from a question to an answer, ScholarPlan goes from a research goal to a plan, retrieves and processes evidence, verifies the results, and can re-plan before producing the final report.

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ScholarPlan uses a multi-agent architecture in which each component has a specialised responsibility. The Planner creates research tasks, the Retriever searches OpenAlex for academic sources, and the Processor generates claims from the retrieved evidence. The Critic then evaluates these claims in a separate verification stage.
The agents coordinate through a shared SQLite Blackboard, which stores research tasks, sources, claims, feedback and execution events. This allows the components to share persistent, run-specific state without communicating directly with one another.
Finally, the Archivist collects the results and generates the research report, bibliography and execution trace.
The main architectural principle is separation of responsibilities combined with shared state and traceable execution.

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The agent begins with a research goal, which is converted into executable tasks. It retrieves academic sources, processes them into claims and evaluates the available evidence.
The important part is what happens next. The system uses verification feedback to decide whether further research is needed. When re-planning is triggered, the Planner creates a revised plan and the next iteration performs another retrieval and verification cycle.
This is a bounded ReAct-style workflow rather than a fixed sequence of actions. Execution is limited to a maximum of three iterations to control resource usage. When the stopping criteria are satisfied, the system archives its results.

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The system was implemented in Python, with Azure AI providing the language model used for planning, clai m generation and verification. OpenAlex serves as the academic retrieval source, while SQLite implements t he shared Blackboard and persists the state of each research run.
I also integrated Sentence Transformers to generate 384-dimensional embeddings using the all-MiniLM-L6-v2 model. These embeddings are stored in SQLite, and NumPy is used to calculate cosine similarity for ident ifying potentially similar sources.
The system was built with separate modules for each agent, which makes the implementation easier to test and maintain. The final outputs include a research report, a BibTeX bibliography, an execution trace and qua ntitative evaluation metrics.

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Here we can see ScholarPlan running with the goal of investigating the impact of large language models on academic research.

The execution completed successfully after two iterations, retrieving 40 academic sources and generating 19 claims. Of these, 17 were classified as supported and two as irrelevant. No claims were classified as unsupported or as having insufficient evidence in this run.
The system also completed the archival stage, producing the research report, bibliography and execution trace. These results provide a concrete example of the complete application running from an initial research goal to a final research package.

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One of the key features of ScholarPlan is its ability to adapt when the available evidence needs further investigation. After the first retrieval and processing cycle, the Critic evaluates the generated claims and records its verification feedback.
The Planner then uses that feedback to generate a revised research plan, including new tasks and search queries. In the execution trace, Event 539 records the start of re-planning, and Event 540 confirms that the new plan was based on feedback.
The system then performs another retrieval, processing and verification cycle. This demonstrates feedback-driven autonomous behaviour rather than simply repeating the same fixed sequence. The process remains bounded by the maximum iteration limit."

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To assess the implementation, I ran unit tests, functional tests and a complete end-to-end execution. All 24 tests passed, providing evidence that the tested components and workflows behaved as expected.
I also evaluated the generated research outputs. The system achieved 100% embedding coverage, meaning all 40 retrieved sources received embeddings. The Critic-supported claim rate was 89.5%, based on 17 supported claims out of 19.
These metrics measure different aspects of the system: test results assess software behaviour, embedding coverage measures processing completeness, and the supported claim rate reflects the Critic's evaluation. None of them alone establishes the factual accuracy of the research. Independent human assessment would be needed to evaluate that more reliably.

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Although the results demonstrate that ScholarPlan runs successfully, there are important limitations to consider.
One strength is its modular architecture. Each agent has a specialised responsibility, while the SQLite Blackboard provides shared state and makes the execution traceable. The system can also use verification feedback to revise its plan rather than following a fixed sequence.
However, the Critic is itself an AI component, so its assessment does not guarantee that a claim is factually correct. The supported claim rate should therefore be interpreted as an internal evaluation measure rather than an objective measure of truth.
Another limitation is that academic retrieval currently relies on OpenAlex, which restricts the range of sources available. The bounded iteration limit also means that some research questions may require more retrieval and verification cycles than the system permits.
Future work could integrate additional academic databases, compare the Critic's decisions with human assessments, and evaluate retrieval relevance and duplicate detection using more systematic benchmarks.
Overall, ScholarPlan demonstrates a functional autonomous research workflow, but further evaluation is needed before its outputs could be considered reliable without human review.

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To conclude, ScholarPlan demonstrates how an LLM-powered agent can go beyond a single prompt and answer by carrying out a structured research workflow.
It combines autonomous planning, academic retrieval, claim processing, evidence verification and feedback-driven re-planning. The SQLite Blackboard coordinates the agents and maintains a traceable record of execution, while the system produces a research report, references and evaluation outputs.
The project successfully completed its end-to-end workflow and passed all 24 tests. However, the results also highlight the importance of independent verification, because an AI critic cannot guarantee factual correctness.
The main lesson from this project is that useful research agents need more than language generation. They also need structured coordination, access to evidence, evaluation mechanisms and transparent execution.
Thank you for listening.